

Tuning the TTGO unit Sketch for your specific deployment

First, this system can run in one of two different modes. Home or Away mode. To set which mode you are going to use for the system, open the Arduino IDE, CIV_Template.ino and at the top of the sketch you will see a line with "int Home = x". Where x is set to either 0 for away mode or 1 for home mode. Now, you may be asking what the difference is between these modes. So go over the z_userprog.ino sketch and at the top of it is a description of the updates that I've made (KK7PZE) and in it there is an explanation of what the modes do. The reason I created this capability is I have one of these units that lives at my home. That means I always use the same antenna, have the same power output requirements and have the same noise floor. So I can adjust the sketch for that fixed environment. But I have another one of these units that is in my portable operations kit for doing things like POTA. So, in Away mode, I can have the power output set differently than home based on my portable antenna, the amount of power I want to use based on my battery needs, disable the RF Gain and Pre-Amp features since each remote location may be different so I'll adjust them from the radio rather than have this unit do it automagically. So set the Home = to 0 or 1 and then compile and upload the sketch and it will run in that mode for you. You could have a single unit and have it set to Home = 1 at the QTH and then just change the Home = 0 and load that into the unit and off you go for portable operations.

Next, let's talk about a bunch of other things in the z_userprog.ino sketch. First thing is to search for "Table to build RF Power and RF Gain per band". In this table you will learn that to set the percentage of TX Power Output or set an RF Gain value that you actually need to set to number. It's strange I know but if you use the chart it's simple. Let say you want to set a value of 27% for something like Power or Gain. Look in the chart at the 3rd column to find the 27%. Then look at the first two columns next to that 27% and you'll see column 1 contains a value of 0 and the second column contains a value of 104.

So how does this relate to what we care about. First, when running in Home = 1 mode or Home mode, you will want to adjust the TX Power Output of the IC-705 for each band to drive the XPA125B amplifier with the amount of output that you want without overdriving and damaging the amp. So, set your band/frequency and set the mode of operation. I personally like to use RTTY for this testing. Turn down the TX Power Output on the radio and key up. Adjust the TX Power Output to the value you want and note it. Use the table we talked about and get the two values that make up the percentage you want. Do this for each band and then it's time to fill out the chart in the z_userprog.ino sketch. Again, we had selected home as the devices mode so now search for "RF Power Per Band Byte 1 when Home = 1 (Home Mode)". There you will see a few lines of code but the important parts are to look for the table for BandPower1 and BandPower2. You'll see a line below each of these describing which band the values are for. So

lets go through each band and let's say the 27% was for 40 meters. Look for 40m in BandPower1 and BandPower2 and set BandPower1's 40m value to 0 and set BandPower2's 40m value to 104. Now after uploading the sketch to the TTGO your device will output 27% on 40 meters. Again, enter the values for each band. If your antenna doesn't support a band then put 0 and 0 to have the radio put out almost no power. You'll also see a note area below these two BandPower where you can put a note of what percentage is really in each one.

If you look below just a bit further you will see a BandPower3 and BandPower4. This is the TX Output Power for when the device is in Home = 0 or Away mode. I generally use lower percentages here to save on battery. Also, I'd recommend testing what the values need to be based on the antenna you use in the field.

Next let's go a but further down and you'll see RfGain1 and RfGain2. This is to set the RF Gain on the 705 per band when running in Home = 1 or Home mode. It uses the same table to get the two values needed to make a percentage.

Now, just a bit further down you will see a section for PreAmp settings. Again, this will only function when in Home = 1 or Home mode. You can set the Pre-Amp of the radio per band to 0=off, Pre-Amp=1 or Pre-Amp=2. Pre-Amp=2 is only available for HF bands. VHF and UHF only support 0(off) or 1(on)

Now, scroll down past the big percentage chart and find the section "RF Power to be used when connected to a hotspot". This section is only used if you have a HotSpot. Say a Pi-Star or WPSD setup for D-Star. You will see that you can enter the RX frequency and TX Frequency that your hotspot is configured for and then using the percentage chart we learned all about earlier you can set what the TX Output Power is when communicating to the HotSpot. I added this so you don't necessarily blast too much power towards the hotspot if it were sitting very close.

Next is something you shouldn't have to modify but just in case, here it is. We know from the Xeigu XPA125B what voltages are required for each band to do the automatic band switching. The design of this solution puts out that power but different components you use in the build and variability of TTGO designs and the length of the data cable to the amp and the amp itself could all change what voltage actually makes it into the amp. So if you search for "Band Switching Voltages per Band" you will find a section that defines the voltages for each band and for each band a correction factor or corrFact. This is an offset value that is + or - that you can use to fine tune the voltages. What I did with my solution was to clamp on test leads to a multi-meter where the data cable plugs into the custom PCB board. I connect to Ground and Band. I connected the data cable to the Amp and then recorded the voltages and I switched from band to band. The I adjusted the corrFact number to dial in the voltage to perfection. Remember

that making changes in the script doesn't do anything until you compile and upload it to the TTGO.

Last two things on this hints and tricks document are these: 1) When the device is set to Home = 1, or Home mode there are two important buttons. Both of these are the ones to the right of the LCD. The upper one is active by default and does the auto RF Gain and Pre-Amp settings we discussed earlier. Pressing the lower button disables that and you have to adjust the RF Gain and Pre-Amp settings. When the device is set to Home = 0, or Away mode it will default as having the lower button pressed.

2) There is a button on the top of the TTGO board. It's the reset/reboot button. If things get weird with your system you can hit that button and after a short period of time your IC-705 will say the device is disconnected. Just reconnect and get back to work. I only have to do this during testing of features and stuff and have never had to do this when just using the device. It's been very stable for me